

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ArenaX

Esports Tournament Management Platform

1. Introduction

1.1 Purpose

The Esports Tournament Management Platform is a web-based application developed to simplify the process of organizing and managing esports tournaments. The system allows players to create teams, register for tournaments, make payments for paid tournaments, view match schedules, and track rankings. Organizers can create tournaments, manage registrations, schedule matches, and update match results.

1.2 Scope

The system provides a centralized platform for managing esports tournaments. It includes user registration, team management, tournament management, payment processing, match scheduling, result management, and leaderboard generation.

1.3 Objectives

- To automate esports tournament management.
 - To reduce manual effort in handling registrations and match results.
 - To provide a centralized platform for players and organizers.
 - To support both free and paid tournaments.
 - To maintain team rankings and performance statistics.
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2. System Overview

The Esports Tournament Management Platform consists of the following major modules:

User and Team Management

Players can register, log in, create teams, and manage team details.

Tournament Management

Organizers can create tournaments, manage registrations, and schedule matches.

Payment Management

Players can register for paid tournaments through a payment system and track payment status.

Match Management

Organizers can create matches, update scores, and declare winners.

Rankings and Statistics

The system automatically updates rankings and displays team performance statistics.

3. Functional Requirements

3.1 User Registration and Login

- Users can create an account.
- Users can log in using valid credentials.
- Users can update their profile information.

3.2 Team Management

- Create a team.
- View team details.
- Manage team members.
- Register teams for tournaments.

3.3 Tournament Management

- Create tournaments.
- Edit tournament details.
- View tournament information.
- Manage participating teams.

3.4 Payment Management

- Support free tournaments.
- Support paid tournaments.
- Process tournament registration payments.
- Store payment status.
- Display payment history.

3.5 Match Management

- Schedule matches.
- Enter match results.
- Update match status.
- Declare match winners.

3.6 Leaderboard Management

- Display team rankings.
 - Calculate points automatically.
 - Update wins and losses.
 - Display team statistics.
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4. Non-Functional Requirements

4.1 Performance Requirements

- Fast response time.
- Support multiple users simultaneously.
- Efficient database operations.

4.2 Security Requirements

- Secure login system.
- Password encryption.
- Authorized access to tournament management functions.

4.3 Reliability Requirements

- Accurate storage of tournament data.
- Reliable leaderboard calculations.
- Consistent payment records.

4.4 Usability Requirements

- User-friendly interface.
- Easy navigation between modules.
- Responsive design.

5. Software Requirements

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Spring Boot
- Spring Data JPA
- Spring Security

Database

- MySQL

Development Tools

- Visual Studio Code
 - IntelliJ IDEA
 - Postman
 - MySQL Workbench
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6. Hardware Requirements

Minimum Requirements

- Processor: Intel i3 or above
- RAM: 4 GB
- Storage: 20 GB Free Space

Recommended Requirements

- Processor: Intel i5 or above

- RAM: 8 GB
 - Storage: 50 GB Free Space
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7. System Modules

Module 1: Authentication Management

- User Registration
- User Login
- Profile Management

Module 2: Team Management

- Team Creation
- Team Listing
- Team Registration for Tournaments

Module 3: Tournament Management

- Tournament Creation
- Tournament Listing
- Tournament Registration

Module 4: Payment Management

- Free Tournament Registration
- Paid Tournament Registration
- Payment Status Tracking

Module 5: Match Management

- Match Scheduling
- Match Result Entry
- Winner Declaration

Module 6: Ranking and Statistics

- Leaderboard Generation
 - Team Ranking
 - Performance Tracking
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8. Database Design

User

- User ID
- Username
- Email
- Password

Team

- Team ID
- Team Name
- Captain ID

Tournament

- Tournament ID
- Tournament Name
- Game Name
- Start Date
- Entry Fee
- Tournament Type

Registration

- Registration ID
- Team ID
- Tournament ID

Payment

- Payment ID
- Team ID
- Tournament ID
- Amount
- Payment Status
- Payment Date

Match

- Match ID
- Team 1
- Team 2
- Score 1

- Score 2
- Winner

Leaderboard

- Team Name
 - Wins
 - Losses
 - Points
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9. User Roles

Player

- Register and log in.
- Create and manage teams.
- Register for tournaments.
- Make payments for paid tournaments.
- View match schedules and rankings.

Organizer

- Create tournaments.
 - Manage tournament registrations.
 - Schedule matches.
 - Update match results.
 - Monitor tournament statistics.
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10. Conclusion

The Esports Tournament Management Platform provides an efficient solution for organizing and managing esports tournaments. The system automates team registration, tournament management, payment handling, match scheduling, result processing, and leaderboard generation. It improves the overall experience for players and organizers while reducing manual effort and increasing operational efficiency.

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